

- Q1.** Find the state equations for the translational mechanical system shown in Figure 1. Determine the order, poles and zeros of the system.

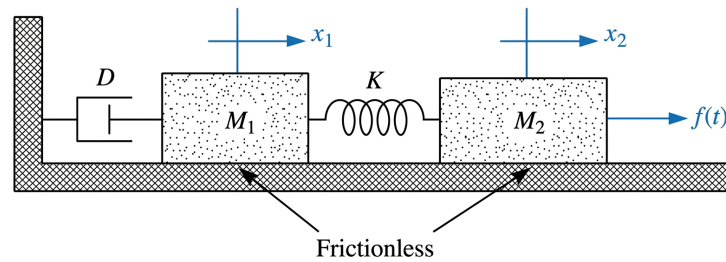


Figure 1: Translational mechanical system

- Q2.** Find the state-space representation of the electrical network shown in Figure 2. Obtain its characteristic equation.

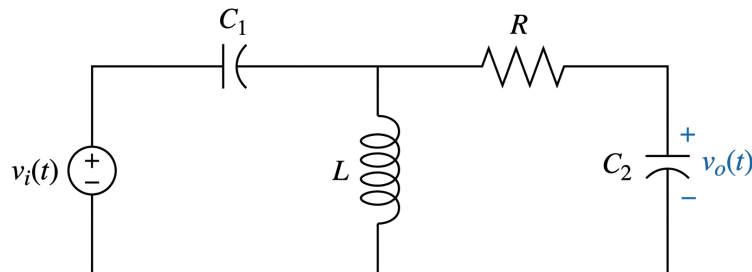


Figure 2: RLC circuit

- Q3.** Convert the following transfer functions into their equivalent state-space representation:

(a) $G(s) = \frac{24}{s^3 + 9s^2 + 26s + 24},$

(b) $G(s) = \frac{s^2 + 7s + 2}{s^3 + 9s^2 + 26s + 24},$

- Q4.** Convert the following state-space model into transfer function; and plot pole and zeros in s-plane:

(a)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u$$

$$y = [1 \quad 0 \quad 0] \mathbf{x}$$

(b)

$$\dot{\mathbf{x}} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u(t)$$

$$y = [1.5 \quad 0.625] \mathbf{x}$$